

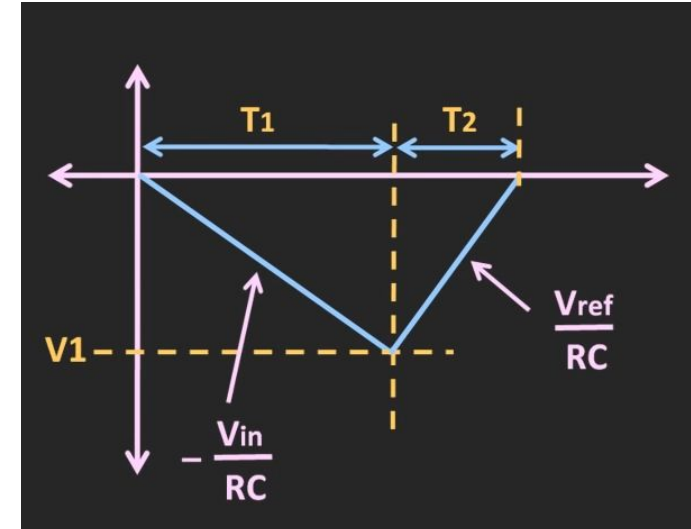
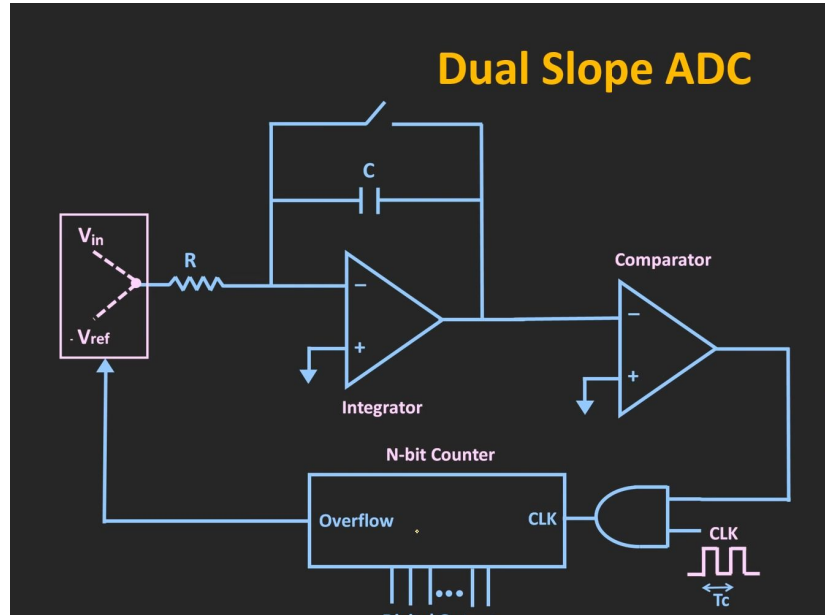
CSE350: Digital Electronics & Pulse Techniques

Lecture 10: Dual Slope ADC , DAC Full

Dual Slope ADC:

Advantages: Requires less components, high resolution (number of quantization levels or bits can be high).

Disadvantages: Too slow conversion.



Operation:

1. Integration Phase (Time T_1)

- The input voltage V_{in} is connected to the integrator for a fixed time T_1 .
- During this phase, the integrator's output voltage (V_1) decreases linearly, as the capacitor integrates the input:

$$V_{out}(t) = -\frac{V_{in}}{RC}t$$

- At the end of T_1 , the integrator output reaches: $V_1 = -\frac{V_{in}T_1}{RC}$

2. De-integration Phase (Time T_2)

- After T_1 , the reference voltage V_{ref} (of opposite polarity to V_{in}) is applied to the integrator.
- The integrator now de-integrates (ramps back up linearly to 0). The slope of the output during de-integration is:

$$\text{Slope} = \frac{V_{ref}}{RC}$$

- The time T_2 taken for the integrator to return to zero is proportional to the input voltage: $T_2 = \frac{V_{in}T_1}{V_{ref}}$

3. Digital Conversion

- A counter operates during T_2 , counting clock pulses (CLK).
- The number of clock cycles during T_2 gives the digital equivalent of V_{in} : $N = f_{clk} \cdot T_2 = f_{clk} \cdot \frac{V_{in} T_1}{V_{ref}}$
- Rearranging:
$$V_{in} = \frac{N \cdot V_{ref}}{f_{clk} \cdot T_1}$$

Key Graph Explanation

- Integrator Output:
 - During T_1 : V_{out} ramps down linearly.
 - During T_2 : V_{out} ramps up linearly until it hits 0.
- The T_2 time is proportional to the input voltage, ensuring precise measurement independent of component tolerances (e.g., R, C).

Operation of the Comparator

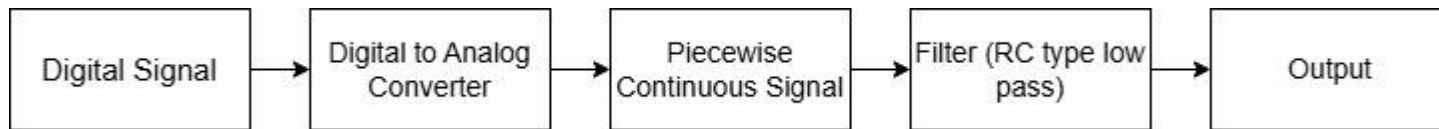
- The comparator has two inputs:
 - Non-inverting Input (+): This is usually tied to a reference voltage, which in this case is 0 V.
 - Inverting Input (-): Connected to the output of the integrator.
- Output Logic:
 - If $V_{\text{out(integrator)}} < 0$: The comparator output is HIGH.
 - If $V_{\text{out(integrator)}} \geq 0$: The comparator output transitions to LOW.
- This transition marks the end of the de-integration phase.

Clock and AND Gate Functionality

- Clock (CLK):
 - A high-frequency clock provides pulses that are counted during the de-integration phase.
 - Each pulse represents a discrete time unit, directly linked to the input voltage V_{in} .

- **AND Gate:**
 - The comparator's output is connected to one input of the AND gate, and the clock signal to the other.
 - As long as the comparator output is HIGH (indicating $V_{\text{out(integrator)}} < 0$), the AND gate allows clock pulses to pass through to the counter.
 - Once the comparator output goes LOW ($V_{\text{out(integrator)}} \geq 0$), the AND gate blocks the clock signal, stopping the counting.

DAC: The process to convert binary bit streams into continuous analog signals. It can be summarized as follows:



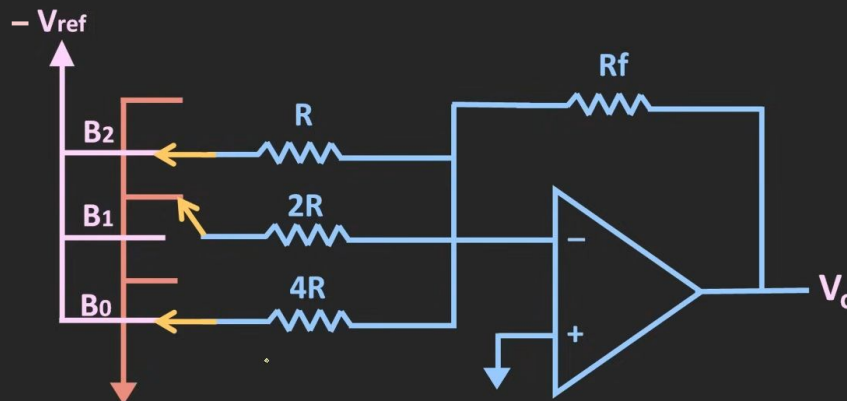
Binary Weighted Resistor

An inverting adder with a feedback resistor $R_f = R$ is used, along with doubling resistors at inverting input terminal. A switch is connected to all branches of input, which may connect to a reference voltage or ground. Since, no current flows through the input terminal of an op-amp and the non-inverting & inverting terminals are virtually shorted, the current towards & voltage at the inverting terminal are both 0.

Let, the current from output to the terminal through R_f is I . Then,

$$I = \frac{V_o - 0}{R_f} = I_2 + I_1 + I_0$$

Binary Weighted Resistor DAC



Inverting Summing Amplifier

Where, I_2 , I_1 & I_0 are the currents through the input branches. When the switches are connected to reference voltage, we can express these currents as,

$$I_2 = \frac{0 - (-V_{ref})}{R}$$

$$I_1 = \frac{0 - (-V_{ref})}{2R}$$

$$I_0 = \frac{0 - (-V_{ref})}{4R}$$

When the switches are connected to ground, the currents are 0. The switches are actually implementations of the bits of the digital input signal. These bits if assumed to be B_2 , B_1 , B_0 , then the currents can be expressed for both cases as,

$$I_2 = B_2 \frac{V_{ref}}{R} \quad I_1 = B_1 \frac{V_{ref}}{2R} \quad I_0 = B_0 \frac{V_{ref}}{4R}$$

Applying KCL at inverting terminal,

$$I = \frac{V_o}{R} = I_2 + I_1 + I_0 = V_{ref} \left(\frac{B_2}{R} + \frac{B_1}{2R} + \frac{B_0}{4R} \right)$$

So,

$$V_o = V_{ref} \left(B_2 + \frac{B_1}{2} + \frac{B_0}{4} \right)$$

The highest output voltage is,

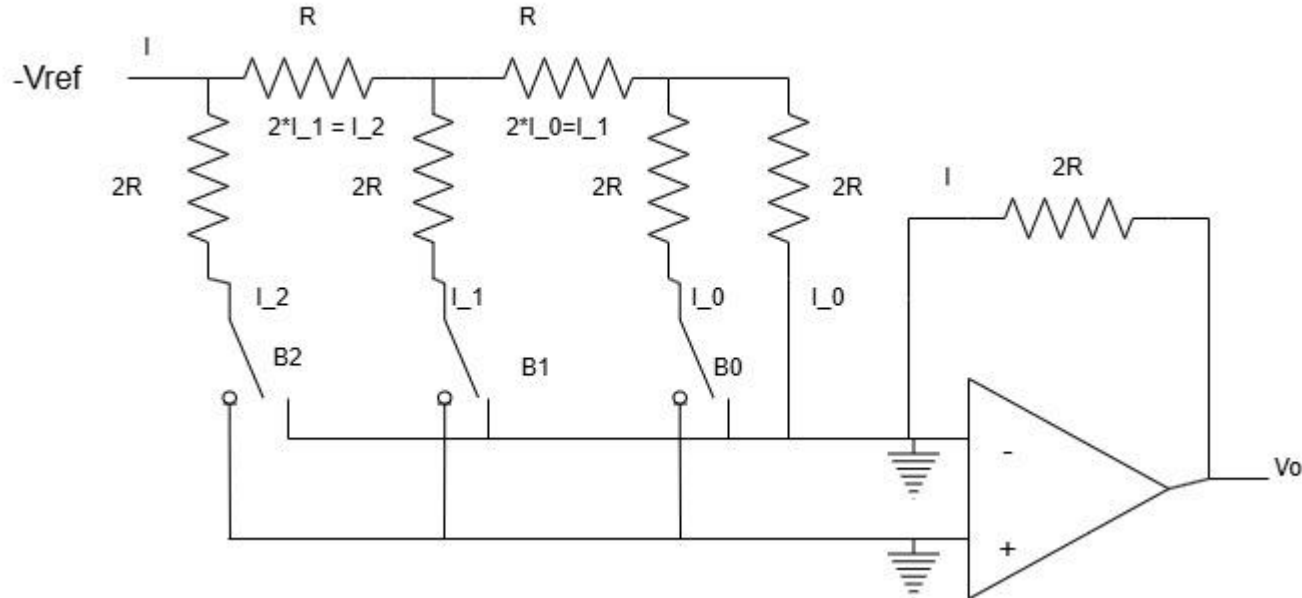
$$V_o = V_{ref} \left(1 + \frac{1}{2} + \frac{1}{4} \right) = V_{ref} \frac{8 - 1}{4} = V_{ref} \frac{2^n - 1}{2^{n-1}}$$

In general, the output voltage can be expressed as,

$$V_o = \left[\frac{-V_{ref} \times R_f}{R} \right] \sum_{i=0}^{i=N-1} \frac{B_i}{2^{(N-1)-i}}$$

One of the disadvantages of Binary Weighted Resistors is that, a lot of resistors are used. Hence, fluctuations in their values can cause errors in output voltage. This sensitive nature can be overcome by the following DAC.

R2R Ladder DAC: Below is shown a 3 bit R2R ladder DAC. Here, as we can see, a resistive network of R s & $2R$ s is used, which has a ladder type structure. The switches connected in $2R$ branches can either connect to the ground at inverting terminal or that at the non-inverting terminal. In the former case, the current can flow through the feedback resistor $2R$. But when the switch connects to non-inverting terminal, it flows to ground.



The currents along the branches are marked on the figure. We get, $I = 2I_2 = 2 * 2I_1 = 2 * 2 * I_0 = 8I_0$

$$\text{So, } I_0 = \frac{I_2}{4} \quad I_1 = \frac{I_2}{2}$$

$$I = I_2 + \frac{I_2}{2} + \frac{I_2}{4}$$

$$\text{Also, } I_2 = \frac{0 - (-V_{ref})}{2R} = \frac{V_{ref}}{2R}$$

Thus, the current through the feedback resistor $2R$ can be expressed using KCL at inverting terminal,

$$I = B_2 \frac{V_{ref}}{2R} + B_1 \frac{V_{ref}}{4R} + B_0 \frac{V_{ref}}{8R} = \frac{V_o - 0}{2R}$$

So, we get,

$$V_o = V_{ref} \left(B_2 + \frac{B_1}{2} + \frac{B_0}{4} \right)$$